

Literature-Implied Answers to Two Ideal-Lattice Questions for $C([0, \omega_1])$

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2026-06-23

Status. Literature-implied answer. The closing remark of Kania–Laustsen [1] asks two questions about the lattice of closed ideals of $\mathcal{B}(C([0, \omega_1]))$. Later work of Kania–Laustsen [2] answers them after identifying the factor space used in the first question and reading the improved lattice diagram in the isomorphic $C_0[0, \omega_1)$ model.

Source questions. In the final remark of [1], after constructing an Eberlein compact space K , the authors prove

$$\mathcal{K}(C([0, \omega_1])) + \overline{\mathcal{G}}_{c_0(\omega_1)}(C([0, \omega_1])) \subsetneq \overline{\mathcal{G}}_{C(K)}(C([0, \omega_1])) \subseteq \mathcal{M}$$

and say that they do not know whether the final inclusion is proper, adding that they conjecture it is. They then ask whether

$$\overline{\mathcal{G}}_{C(K_\alpha) \oplus c_0(\omega_1)}(C([0, \omega_1])) \subseteq \mathcal{SL}_{\alpha+1}(C([0, \omega_1]))$$

is proper for some, or each, countable ordinal α , where $K_\alpha = [0, \omega^{\omega^\alpha}]$.

Identification of the first factor space. The compact space K in [1] is the one-point compactification of the disjoint union of $K_\alpha = [0, \omega^{\omega^\alpha}]$, $\alpha < \omega_1$. Hence

$$C(K) \cong \mathbb{K} \oplus \left(\bigoplus_{\alpha < \omega_1} C(K_\alpha) \right)_{c_0}.$$

Let $A = \{\omega^{\omega^\alpha} : \alpha < \omega_1\} \subset [0, \omega_1)$. The non-scalar summand is exactly $E_A = (\oplus_{\gamma \in A} C[0, \gamma])_{c_0}$ in the notation of [2]. Lemma 2.4 of [2] states that $C_0(L_0)$ is isometric to E_{ω_1} , and that $E_A \cong E_{\omega_1}$ whenever A is uncountable. The same lemma also gives enough absorption to remove the scalar summand: $E_{\omega_1} \cong \mathbb{K} \oplus E_{\omega_1}$. Thus

$$C(K) \cong C_0(L_0).$$

Answer to the conjectured proper inclusion. The later paper records, using the preceding work [1, 3], that the Loy–Willis ideal on $C_0[0, \omega_1)$ is precisely the ideal of operators factoring through $C_0(L_0)$:

$$\mathcal{M} = \mathcal{G}_{C_0(L_0)}.$$

It also notes that the Banach algebras $\mathcal{B}(C_0[0, \omega_1))$ and $\mathcal{B}(C[0, \omega_1])$ have isomorphic lattices of closed ideals because $C_0[0, \omega_1) \cong C[0, \omega_1]$. Combining these facts with $C(K) \cong C_0(L_0)$, the source inclusion

$$\overline{\mathcal{G}}_{C(K)}(C([0, \omega_1])) \subseteq \mathcal{M}$$

is an equality. Therefore the conjecture in [1] that the inclusion is proper is false.

Answer to the Szlenk-inclusion question. For every countable ordinal $\alpha \geq 1$, [2] proves

$$\overline{\mathcal{G}}_{C(K_\alpha) \oplus c_0(\omega_1)}(C_0[0, \omega_1)) \subsetneq \overline{\mathcal{G}}_{c_0(\omega_1, C(K_\alpha))}(C_0[0, \omega_1)) \subseteq \mathcal{SZ}_{\alpha+1}(C_0[0, \omega_1)).$$

Transferring through the isomorphism $C_0[0, \omega_1) \cong C([0, \omega_1])$, this proves that the inclusion asked about in [1] is proper for every nonzero countable ordinal α . This packet does not claim the $\alpha = 0$ endpoint. The stronger question left open in [2], whether $\overline{\mathcal{G}}_{c_0(\omega_1, C(K_\alpha))}$ equals $\mathcal{SZ}_{\alpha+1}$, is also not claimed.

Provenance and novelty check. The cheap run indexes had no exact entry for arXiv:1112.4800 or the two closing ideal-lattice questions. Web searches for exact phrases involving “Loy–Willis ideal”, “ $C([0, \omega_1])$ ”, “closed ideals”, and the properness question mainly rediscovered the source paper. The answer above is therefore recorded as a literature-implied identification, not as an original new proof.

References

- [1] T. Kania and N. J. Laustsen, *Uniqueness of the maximal ideal of the Banach algebra of bounded operators on $C([0, \omega_1])$* , arXiv:1112.4800.
- [2] T. Kania and N. J. Laustsen, *Operators on two Banach spaces of continuous functions on locally compact spaces of ordinals*, arXiv:1304.4951.
- [3] T. Kania, P. Koszmider, and N. J. Laustsen, the work cited as [3] in [2], specifically the theorem characterizing the Loy–Willis ideal as factoring through $C_0(L_0)$.